17. Results and Findings

Raw-file findings

Finding 1: Broad coverage across categories and geography

The dataset has 100,000 rows, 15 vehicle categories, 200 manufacturer labels, and 7 market regions. The most common category is **Two-Wheeler Scooter** (9,826 records; 9.83%). This describes dataset composition only.

Finding 2: Price is right-skewed

Average recorded price is $65,010.46, compared with a median of $42,808.69. The maximum is $409,995.49. Because the mean exceeds the median and the upper tail is wide, use medians/box plots as well as averages.

Finding 3: Range distribution

Average range is 324.31 km and median range is 338.30 km. The observed range runs from 55.0 to 659.9 km. Category comparisons should include category counts and use consistent units.

Finding 4: Price and battery capacity are associated

Pearson correlation between `price_usd` and `battery_capacity_kwh` is **0.837**. This is a strong positive linear association in the supplied file, but it does not show that battery capacity alone causes higher prices; vehicle class and other specifications may also matter.

Finding 5: Annual sales require robust treatment

The median `annual_sales_units` value is 3,935, the mean is 7,803.26, and the maximum is 772,442. The field has 989 missing values. The maximum is an extreme value relative to the median and should be checked against source definitions before using total or mean sales.

Finding 6: Data completeness is mostly high but not uniform

The raw file contains 9,511 missing cells (0.257% of all cells). Missingness is concentrated in cycle life, resale value, annual sales, and customer rating. No duplicate complete rows or duplicate vehicle IDs were detected.

Limitations on findings

The data source and generation method are not provided. These findings cannot independently establish global EV market share, actual sales performance, emissions reduction, or real-world vehicle performance. Any public-facing claims should be restricted to observed dataset values until provenance is verified.